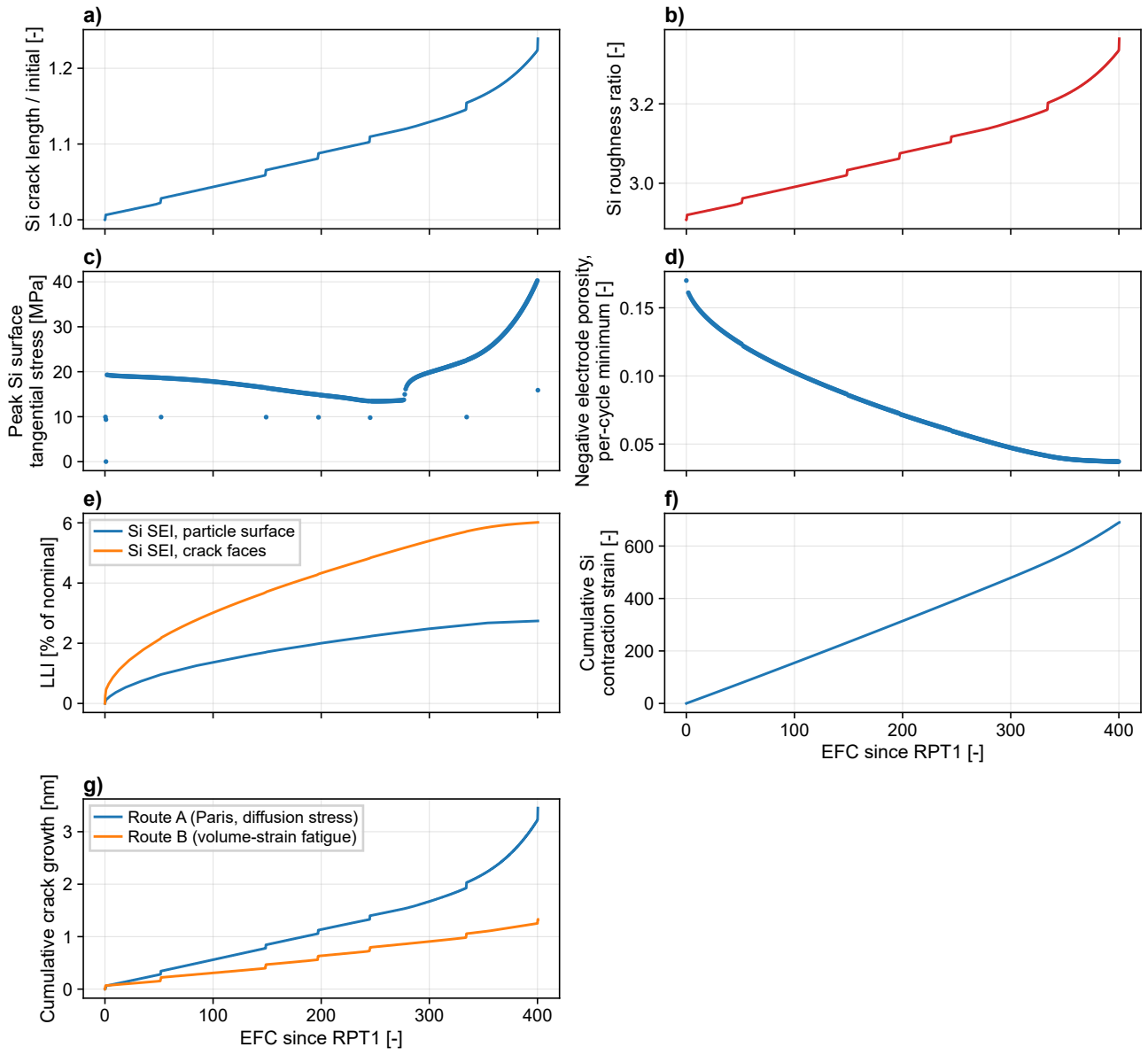

a)
Si crack length / initial [-]
1.0
1.1
1.2
b)
Si roughness ratio [-]
3.0
3.2
c)
Peak Si surface tangential stress [MPa]
0
10
20
30
40
d)
Negative electrode porosity, per-cycle minimum [-]
0.05
0.10
0.15
e)
LLI [% of nominal]
0
2
4
6
Si SEI, particle surface
Si SEI, crack faces
f)
Cumulative Si contraction strain [-]
0
200
400
600
0
100
200
300
400
EFC since RPT1 [-]
g)
Cumulative crack growth [nm]
0
1
2
3
Route A (Paris, diffusion stress)
Route B (volume-strain fatigue)
0
100
200
300
400
EFC since RPT1 [-]